Quadrature Rotary Encoder

Make every Gray-code edge and invalid jump visible

Count contact transitions, not vendor-dependent clicks

Lesson	032 — component	**Time**	120–180 minutes
Prerequisites	Lessons 001, 002, 006, 016	**Board**	Arduino Mega 2560
Evidence	TP-A, TP-B, position nibble, RGB state	**Status**	host verified; bench open
Energy class	E1: USB, passive contacts, resistor-limited LEDs		

Specimen gate. The Elegoo listing authorizes the rotary encoder product family, not a universal module pin order or electrical revision. Identify the exact markings and pins before wiring. Do not infer `CLK`, `DT`, switch, supply, or ground order from a photograph or retail family name.

1 Mission and learning outcomes

$$\text{phase A} + \text{phase B} \rightarrow \text{Gray-code edge} \rightarrow \text{delta} + \text{position} + \text{diagnostics}.$$

The unit is one valid edge, not a mechanical detent or "click." By the end, you can:

- trace both valid Gray-code directions from any starting phase;
- distinguish repeated states, bounce reversals, and invalid jumps;
- explain why the decoder establishes a baseline without movement;
- verify transactional acquisition and reverse-order shutdown;
- separate electrical phase evidence from software presentation;
- record host evidence without claiming a physical result.

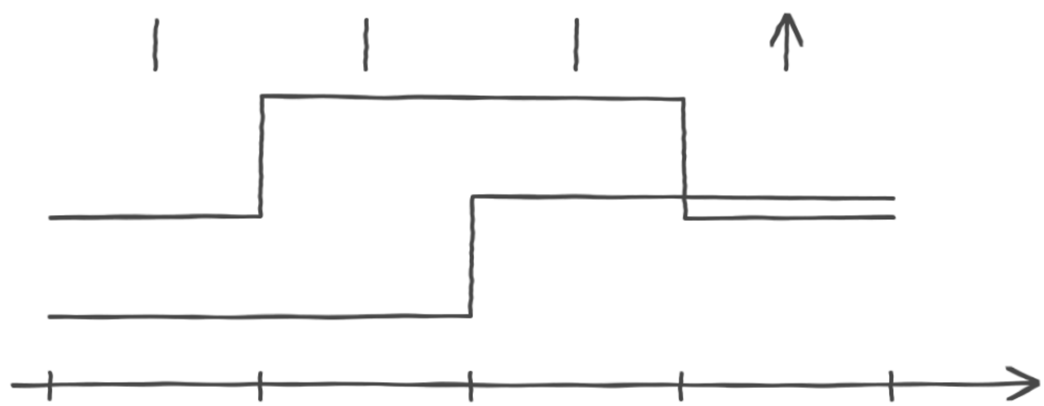
2 Literal wiring and unpowered inspection

Role	Mega pin	Literal path
phase A / TP-A	D24	identified `CLK/A` contact to D24; internal pull-up
phase B / TP-B	D25	identified `DT/B` contact to D25; internal pull-up
separate switch	D26	identified `SW` contact to D26; separate `Button`
position nibble	D30–D33	pin to 1 kΩ, LED anode; cathode to GND
RGB decision state	D5–D7	each channel through its own 1 kΩ resistor
reference	GND	identified module contact ground to Mega GND

The encoder component owns only D24 and D25. D26 remains compositionally separate so the same decoder works with encoders that have no push switch. The four nibble LEDs show the low position bits. The RGB LED pulses green for a valid edge, remains amber for no edge, and pulses red for an invalid jump. Cadence and the nibble make the evidence interpretable without color alone.

Unpowered boundary. The exact module markings, revision, ground contact, supply rating, and polarity must already be resolved. Continuity must associate D24 and D25 with the intended contacts and D26 with the separate switch path. Every LED channel retains its own resistor.

3 Gray-code timing drawing



Upper trace: **A**. Lower trace: **B**. Samples: **00** → **01** → **11** → **10** → **00**. Four valid clockwise edges.

Alternative text: phase B rises, phase A rises, phase B falls, then phase A falls, producing the sampled masks 00, 01, 11, 10, 00 and four clockwise edge decisions. Reverse traversal reports counterclockwise.

Before power, the expected TP-A and TP-B order is the printed Gray-code sequence. A different observed order is evidence to stop and reconcile the fixture, not a value to reinterpret silently.

4 Decoder contract

Observation	Deterministic result
valid clockwise edge	`delta = +1`; increment position
valid counterclockwise edge	`delta = -1`; decrement position
same phase	zero delta; preserve position and invalid count
two-bit jump	zero delta; increment invalid count; adopt new baseline
position limit	hold at `INT32_MIN/MAX`; report saturation

`reversed` exchanges the reported signs. Bounce reversals are decoded literally; the component does not guess a detent. The invalid-transition count saturates at 65,535. Position saturates at its signed 32-bit limits and never wraps. `resetPosition()` changes only accumulated position and clears position saturation; it preserves the current hardware baseline.

Initialization validates distinct Mega digital pins before claims, acquires A then B, configures the selected pull, and reads A then B exactly once to set a movement-free baseline. Each update again reads A once and then B once.

5 Lifecycle and failure evidence

If B acquisition or configuration fails, initialization rolls A back. Repeated successful initialization performs no new hardware work. Shutdown releases B then A, is idempotent, and retains count evidence while the snapshot status becomes `NotInitialized`. Destruction follows the same cleanup. Copy and move operations are prohibited.

Evidence question	Required proof
Did ownership succeed?	ready indication only after both phase inputs initialize
Did decoding succeed?	TP sequence agrees with delta and position nibble
Was a jump rejected?	unchanged position plus incremented count and red pulse
Did shutdown become safe?	RGB/nibble inactive and both input modes released

An inactive LED does not prove input release. Record acquisition and post-shutdown input-mode evidence separately.

6 Narrative Mega example

```
observeEncoder();
decideEncoderEvidence();
presentEncoderEvidence();
```

Setup acquires phase inputs, the separate button, nibble LEDs, and RGB indicator; configures every output inactive; initializes the decoder; then shows ready. The loop observes exactly one phase pair, decides from one stable snapshot, and presents its low nibble and diagnostic state.

7 Predict, observe, interpret

1. Predict the complete TP-A/TP-B sequence before applying USB power.
2. Rotate slowly through one full electrical sequence. Observe TP-A and TP-B, then observe four signed edges independent of mechanical detents.
3. Compare the electrical sequence, low position nibble, and valid-edge pulse. Interpret agreement as decoder evidence.
4. Reverse direction and deliberately include a small bounce reversal. Predict literal opposing edges rather than hidden filtering.
5. Use only an unpowered, pre-inspected contact fixture to arrange a safe two-bit jump. Predict unchanged position, one invalid increment, and a red pulse after power is restored.
6. Invoke shutdown separately. Record inactive displays and input release, then remove USB power.

8 Diagnosis and stop conditions

Observation	Action and interpretation
count direction reversed	verify A/B identity; use configuration reversal only intentionally
two bits change together	inspect contacts and sampling; expect invalid evidence
LED count disagrees with TP sequence	stop and inspect mapping before proceeding
position changes on an invalid jump	software evidence has failed; do not accept
hot part, resets, unstable or out-of-rail TP	remove USB immediately
unidentified module contact	remain unpowered until primary evidence identifies it

Never move jumpers while powered. Software shutdown is lifecycle behavior, not an emergency stop; remove USB power for any unsafe or unexplained condition.

9 The two-minute encoder adventure

The RGB witness makes a dependent three-step story. Turn through a positive edge to unlock the first stage, turn through a negative edge to unlock the second, then press the encoder switch to reset the count and reach the steady white finish. The four position lamps continue to expose the low nibble throughout. At 120 seconds the sketch shuts every indicator down regardless of contact state or progress.

10 Exercises

1. Starting at 11, list the next masks and deltas for one reverse cycle.
2. Explain why adopting the new phase after an invalid jump enables recovery without inventing movement.
3. Predict the nibble at positions -1, 0, 7, 15, and 16.
4. Design a trace that distinguishes a repeated sample from contact bounce.
5. Explain why the separate switch belongs to `Button`, not the quadrature decoder.

11 Software evidence

Tests cover all starting phases, directions, reversal, partial turns, bounce, repeated states, every invalid jump and recovery, both saturations, reset, exact read order, transactional failures, lifecycle, traits, and replay. Repository automation runs those tests, Mega compilation, size, and publication gates. They are supporting evidence, not learner chores or physical acceptance.

12 Reference trail

- ADK `quadrature_encoder.h`, focused tests, and canonical Lesson 032 sketch;
- ADK Lessons 031–033 design record and authorized Elegoo family ledger;
- Arduino Mega 2560 Rev3 datasheet for the exact board profile.

The searchable HTML reference is the primary accessible route. This printable PDF uses real text, headings, lists, and a monochrome timing drawing but does not claim PDF/UA conformance.